

Lecture 08

Concavity and Points of Inflection

Taylor and Maclaurin Series

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1 Concavity

The derivative $f'(x)$ tells us whether a function is increasing or decreasing. The way in which $f'(x)$ itself changes describes the *concavity* of the graph.

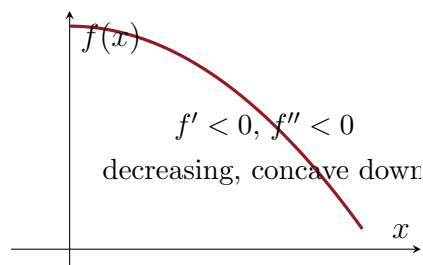
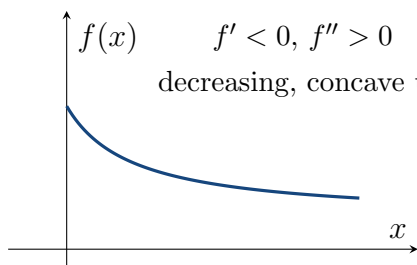
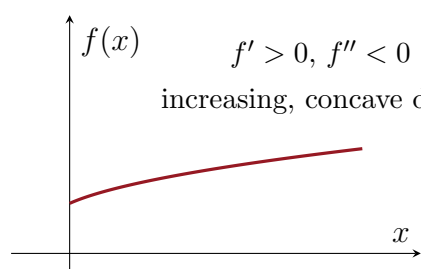
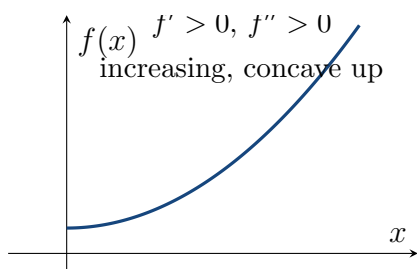
Definition

Let f be differentiable on an open interval I .

- (i) If f' is increasing on I , then f is **concave up** on I .
- (ii) If f' is decreasing on I , then f is **concave down** on I .

Geometrically, a concave-up curve bends like a cup, while a concave-down curve bends like an inverted cup. Concavity does not by itself determine whether the function is increasing or decreasing.

1.1 Four possible combinations



1.2 Second derivative test for concavity

Theorem

Let f be twice differentiable on an interval I .

- (i) If $f''(x) > 0$ for every $x \in I$, then f is concave up on I .
- (ii) If $f''(x) < 0$ for every $x \in I$, then f is concave down on I .

The theorem follows from the fact that f'' is the derivative of f' . Hence, $f'' > 0$ means that f' is increasing, and $f'' < 0$ means that f' is decreasing.

2 Points of Inflection

Definition: Point of Inflection

Suppose that f is continuous at $x = a$. If the graph of f changes concavity at $x = a$, then the point

$$(a, f(a))$$

is called a **point of inflection** of the graph of f .

Possible candidates for inflection points occur where

$$f''(x) = 0 \quad \text{or} \quad f''(x) \text{ is undefined.}$$

However, a candidate becomes an inflection point only when the concavity actually changes across that value.

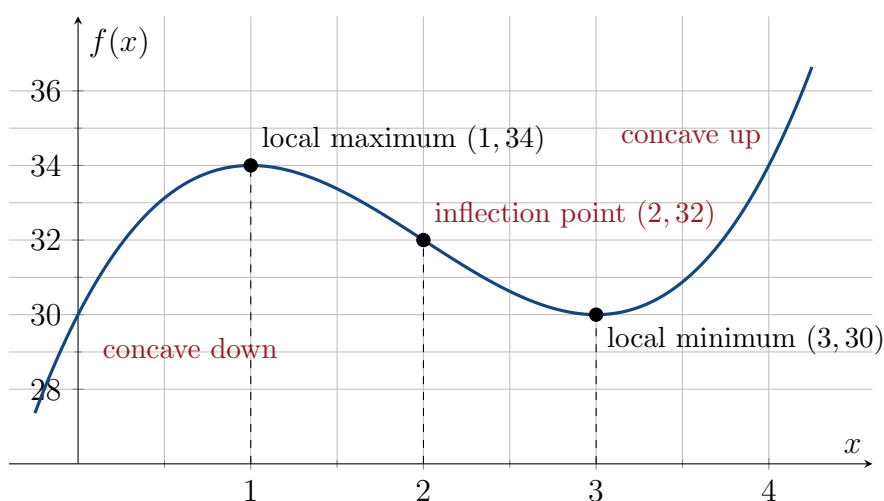
Important Note

The equation $f''(a) = 0$ is not sufficient by itself. For example, for $f(x) = x^4$,

$$f''(x) = 12x^2, \quad f''(0) = 0,$$

but $f''(x) > 0$ on both sides of 0. Therefore, $(0, 0)$ is not an inflection point.

2.1 A graph containing critical and inflection points



3 Worked Example on Concavity and Inflection

Example 1

Let

$$f(x) = x^3 - 6x^2 + 9x + 30.$$

Determine the intervals on which f is concave up and concave down. Find all points of inflection. Also identify the critical points and use the graph to confirm the results.

Answer. We first calculate the derivatives:

$$\begin{aligned}f'(x) &= 3x^2 - 12x + 9 = 3(x-1)(x-3), \\f''(x) &= 6x - 12 = 6(x-2).\end{aligned}$$

Step 1: Critical points

Critical points occur where $f'(x) = 0$:

$$3(x-1)(x-3) = 0 \implies x = 1, 3.$$

Their coordinates are

$$\begin{aligned}f(1) &= 1 - 6 + 9 + 30 = 34, \\f(3) &= 27 - 54 + 27 + 30 = 30.\end{aligned}$$

Thus, the critical points are $(1, 34)$ and $(3, 30)$.

Since f' changes from positive to negative at $x = 1$, $(1, 34)$ is a local maximum. Since f' changes from negative to positive at $x = 3$, $(3, 30)$ is a local minimum.

Step 2: Candidate for an inflection point

Set $f''(x) = 0$:

$$6x - 12 = 0 \implies x = 2.$$

Also,

$$f(2) = 8 - 24 + 18 + 30 = 32.$$

Hence, the candidate point is $(2, 32)$.

Step 3: Concavity test

The value $x = 2$ divides the real line into two intervals.

Interval	Test point	Sign of $f''(x) = 6x - 12$	Conclusion
$(-\infty, 2)$	$x = 0$	$f''(0) = -12 < 0$	f is concave down
$(2, \infty)$	$x = 3$	$f''(3) = 6 > 0$	f is concave up

Therefore,

$$f \text{ is concave down on } (-\infty, 2)$$

and

$$f \text{ is concave up on } (2, \infty).$$

Since the concavity changes at $x = 2$, the point

$$(2, 32)$$

is a point of inflection.

Observation

At an inflection point the graph changes the direction in which it bends. The slope $f'(2) = -3$ is not zero, so an inflection point need not be a critical point.

4 Taylor Series

Taylor series express a sufficiently differentiable function as an infinite power series centred at a specified point.

Definition: Taylor Series

The Taylor series of a function f about $x = a$ is

$$\begin{aligned}
 f(x) &= f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots \\
 &\quad + \frac{f^{(n)}(a)}{n!}(x-a)^n + \cdots \\
 &= \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n,
 \end{aligned}$$

provided that the series converges to $f(x)$.

4.1 Taylor polynomial

When the Taylor series is truncated after the term of degree n , we obtain the Taylor polynomial of degree n about $x = a$:

$$P_{n,a}(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k.$$

That is,

$$P_{n,a}(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n.$$

Meaning of the Taylor Polynomial

The polynomial $P_{n,a}$ has the same value and the same first n derivatives as f at $x = a$:

$$P_{n,a}^{(k)}(a) = f^{(k)}(a), \quad k = 0, 1, \dots, n.$$

Therefore, near $x = a$, the Taylor polynomial can be used to approximate f .

5 Taylor Series of $\cos x$ about $x = \frac{\pi}{2}$

Example 2

Find the Taylor series expansion of

$$f(x) = \cos x$$

about $x = \frac{\pi}{2}$.

Answer. Let $a = \frac{\pi}{2}$. The derivatives repeat in a cycle:

$$\begin{array}{ll} f(x) = \cos x, & f\left(\frac{\pi}{2}\right) = 0, \\ f'(x) = -\sin x, & f'\left(\frac{\pi}{2}\right) = -1, \\ f''(x) = -\cos x, & f''\left(\frac{\pi}{2}\right) = 0, \\ f^{(3)}(x) = \sin x, & f^{(3)}\left(\frac{\pi}{2}\right) = 1, \\ f^{(4)}(x) = \cos x, & f^{(4)}\left(\frac{\pi}{2}\right) = 0. \end{array}$$

Substituting into the Taylor formula,

$$\begin{aligned}\cos x &= 0 - \left(x - \frac{\pi}{2}\right) + 0 \cdot \frac{\left(x - \frac{\pi}{2}\right)^2}{2!} + \frac{\left(x - \frac{\pi}{2}\right)^3}{3!} - \frac{\left(x - \frac{\pi}{2}\right)^5}{5!} + \dots \\ &= -\left(x - \frac{\pi}{2}\right) + \frac{\left(x - \frac{\pi}{2}\right)^3}{3!} - \frac{\left(x - \frac{\pi}{2}\right)^5}{5!} + \dots.\end{aligned}$$

Hence,

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^{n+1}}{(2n+1)!} \left(x - \frac{\pi}{2}\right)^{2n+1}$$

for every real x .

Equivalently, because

$$\cos x = -\sin\left(x - \frac{\pi}{2}\right),$$

the same expansion follows immediately from the Maclaurin series of $\sin x$.

6 Additional Taylor Series Examples

Example 3: Taylor Series of e^x about $x = 1$

Find the Taylor series expansion of

$$f(x) = e^x$$

about $x = 1$.

Answer. Since every derivative of e^x is e^x , we have

$$f^{(n)}(1) = e \quad \text{for every } n \geq 0.$$

Therefore,

$$\begin{aligned}e^x &= e + e(x-1) + \frac{e}{2!}(x-1)^2 + \frac{e}{3!}(x-1)^3 + \dots \\ &= e \left[1 + (x-1) + \frac{(x-1)^2}{2!} + \frac{(x-1)^3}{3!} + \dots \right].\end{aligned}$$

Hence,

$$e^x = e \sum_{n=0}^{\infty} \frac{(x-1)^n}{n!}$$

for every real x .

Example 4: Taylor Series of $\ln x$ about $x = 1$

Find the Taylor series expansion of

$$f(x) = \ln x$$

about $x = 1$.

Answer. We calculate

$$\begin{aligned} f(1) &= 0, \\ f'(x) &= \frac{1}{x}, & f'(1) &= 1, \\ f''(x) &= -\frac{1}{x^2}, & f''(1) &= -1, \\ f^{(3)}(x) &= \frac{2!}{x^3}, & f^{(3)}(1) &= 2!, \\ f^{(4)}(x) &= -\frac{3!}{x^4}, & f^{(4)}(1) &= -3!. \end{aligned}$$

For $n \geq 1$,

$$f^{(n)}(1) = (-1)^{n-1}(n-1)!.$$

Thus,

$$\begin{aligned} \ln x &= (x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3} - \frac{(x-1)^4}{4} + \dots \\ &= \sum_{n=1}^{\infty} (-1)^{n-1} \frac{(x-1)^n}{n}. \end{aligned}$$

Therefore,

$$\boxed{\ln x = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{(x-1)^n}{n}}$$

Example 5: Taylor Series of $e^{5x} \sin x$ about $x = \frac{\pi}{2}$

Find the Taylor series expansion of

$$y = e^{5x} \sin x$$

about $x = \frac{\pi}{2}$ using successive derivatives.

Answer. Let

$$f(x) = e^{5x} \sin x, \quad a = \frac{\pi}{2}.$$

The Taylor series of f about $x = a$ is

$$f(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \frac{f^{(3)}(a)}{3!}(x - a)^3 + \cdots.$$

We now calculate the successive derivatives:

$$\begin{aligned} f(x) &= e^{5x} \sin x, \\ f'(x) &= e^{5x}(5 \sin x + \cos x), \\ f''(x) &= e^{5x}(24 \sin x + 10 \cos x), \\ f^{(3)}(x) &= e^{5x}(110 \sin x + 74 \cos x), \\ f^{(4)}(x) &= e^{5x}(476 \sin x + 480 \cos x), \\ f^{(5)}(x) &= e^{5x}(1900 \sin x + 2876 \cos x), \\ f^{(6)}(x) &= e^{5x}(6624 \sin x + 16280 \cos x). \end{aligned}$$

Since

$$\sin \frac{\pi}{2} = 1, \quad \cos \frac{\pi}{2} = 0,$$

we obtain

$$\begin{aligned} f\left(\frac{\pi}{2}\right) &= e^{5\pi/2}, \\ f'\left(\frac{\pi}{2}\right) &= 5e^{5\pi/2}, \\ f''\left(\frac{\pi}{2}\right) &= 24e^{5\pi/2}, \\ f^{(3)}\left(\frac{\pi}{2}\right) &= 110e^{5\pi/2}, \\ f^{(4)}\left(\frac{\pi}{2}\right) &= 476e^{5\pi/2}, \\ f^{(5)}\left(\frac{\pi}{2}\right) &= 1900e^{5\pi/2}, \\ f^{(6)}\left(\frac{\pi}{2}\right) &= 6624e^{5\pi/2}. \end{aligned}$$

Substituting these values into the Taylor formula gives

$$\begin{aligned} e^{5x} \sin x &= e^{5\pi/2} \left[1 + 5 \left(x - \frac{\pi}{2}\right) + \frac{24}{2!} \left(x - \frac{\pi}{2}\right)^2 + \frac{110}{3!} \left(x - \frac{\pi}{2}\right)^3 \right. \\ &\quad \left. + \frac{476}{4!} \left(x - \frac{\pi}{2}\right)^4 + \frac{1900}{5!} \left(x - \frac{\pi}{2}\right)^5 + \frac{6624}{6!} \left(x - \frac{\pi}{2}\right)^6 + \cdots \right]. \end{aligned}$$

Therefore,

$$e^{5x} \sin x = e^{5\pi/2} \left[1 + 5 \left(x - \frac{\pi}{2} \right) + 12 \left(x - \frac{\pi}{2} \right)^2 + \frac{55}{3} \left(x - \frac{\pi}{2} \right)^3 + \frac{119}{6} \left(x - \frac{\pi}{2} \right)^4 + \frac{95}{6} \left(x - \frac{\pi}{2} \right)^5 + \frac{46}{5} \left(x - \frac{\pi}{2} \right)^6 + \cdots \right].$$

Taylor Series versus Taylor Polynomial

A Taylor *series* contains infinitely many terms, whereas a Taylor polynomial keeps only finitely many terms. For example, the fourth-degree Taylor polynomial for $f(x) = e^{5x} \sin x$ about $x = \frac{\pi}{2}$ is

$$P_4(x) = e^{5\pi/2} \left[1 + 5 \left(x - \frac{\pi}{2} \right) + 12 \left(x - \frac{\pi}{2} \right)^2 + \frac{55}{3} \left(x - \frac{\pi}{2} \right)^3 + \frac{119}{6} \left(x - \frac{\pi}{2} \right)^4 \right].$$

It provides a convenient local approximation when x is close to $\frac{\pi}{2}$.

7 Maclaurin Series

Definition: Maclaurin Series

A Taylor series centred at $a = 0$ is called a **Maclaurin series**. Thus,

$$\begin{aligned} f(x) &= f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f^{(3)}(0)}{3!}x^3 + \cdots \\ &= \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n, \end{aligned}$$

provided the series converges to $f(x)$.

8 Maclaurin Series of $\frac{1}{1-x}$

Example 6

Find the Maclaurin series expansion of

$$f(x) = \frac{1}{1-x}.$$

Answer. We have

$$f(x) = (1-x)^{-1}.$$

Its derivatives are

$$\begin{aligned} f'(x) &= (1-x)^{-2}, \\ f''(x) &= 2(1-x)^{-3}, \\ f^{(3)}(x) &= 3!(1-x)^{-4}, \\ &\vdots \\ f^{(n)}(x) &= n!(1-x)^{-(n+1)}. \end{aligned}$$

Therefore,

$$f^{(n)}(0) = n!.$$

Substituting into the Maclaurin formula gives

$$\begin{aligned} \frac{1}{1-x} &= \sum_{n=0}^{\infty} \frac{n!}{n!} x^n \\ &= 1 + x + x^2 + x^3 + \cdots. \end{aligned}$$

Hence,

$$\boxed{\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n} \quad \text{for } |x| < 1.$$

Convergence Condition

The expansion is the geometric series with common ratio x . It converges only when $|x| < 1$. At $x = 1$ it becomes $1 + 1 + 1 + \cdots$, and at $x = -1$ it becomes $1 - 1 + 1 - 1 + \cdots$; neither converges in the ordinary sense.

9 Additional Maclaurin Series Examples

Example 7: Maclaurin Series of e^x

Find the Maclaurin series of e^x .

Answer. Since $f^{(n)}(x) = e^x$ for every n ,

$$f^{(n)}(0) = 1.$$

Hence,

$$\boxed{e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots = \sum_{n=0}^{\infty} \frac{x^n}{n!}}$$

for every $x \in \mathbb{R}$.

Example 8: Maclaurin Series of $\sin x$ and $\cos x$

Find the Maclaurin series of $\sin x$ and $\cos x$.

Answer. The derivatives of sine and cosine repeat after every four differentiations. At $x = 0$, only alternating odd derivatives of $\sin x$ and alternating even derivatives of $\cos x$ are nonzero. Therefore,

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$$

and

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$$

for every real x .

Example 9: Maclaurin Series of $\ln(1+x)$

Find the Maclaurin series of

$$f(x) = \ln(1+x).$$

Answer. From the Taylor expansion of $\ln x$ about $x = 1$, replace $x - 1$ by x . Thus,

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n}$$

for $-1 < x \leq 1$. At $x = 1$,

$$\ln 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots.$$

Example 10: Obtaining a New Series by Substitution

Use the Maclaurin series of e^x to find a series for e^{x^2} .

Answer. Starting from

$$e^u = \sum_{n=0}^{\infty} \frac{u^n}{n!},$$

put $u = x^2$. Then

$$e^{x^2} = 1 + x^2 + \frac{x^4}{2!} + \frac{x^6}{3!} + \cdots = \sum_{n=0}^{\infty} \frac{x^{2n}}{n!}$$

for every real x .

9.1 Frequently used Maclaurin series

Function	Maclaurin series	Validity
e^x	$\sum_{n=0}^{\infty} \frac{x^n}{n!}$	all x
$\sin x$	$\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$	all x
$\cos x$	$\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$	all x
$\frac{1}{1-x}$	$\sum_{n=0}^{\infty} x^n$	$ x < 1$
$\ln(1+x)$	$\sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n}$	$-1 < x \leq 1$
$\tan^{-1} x$	$\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}$	$ x \leq 1$

Useful Operations on Power Series

Within the interval of convergence, a power series may be differentiated, integrated, multiplied by a power of x , or used with a suitable substitution. These operations often produce new series without repeatedly calculating higher derivatives.

10 Summary

Concept	Main criterion or formula
Concave up	f' is increasing; when f'' exists, usually test $f''(x) > 0$.
Concave down	f' is decreasing; when f'' exists, usually test $f''(x) < 0$.
Inflection point	The graph changes concavity at $x = a$ and f is continuous there.
Candidate values	Solve $f''(x) = 0$ and include values where f'' is undefined.
Taylor series about a	$\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x - a)^n.$
Taylor polynomial	$P_{n,a}(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x - a)^k.$
Maclaurin series	Taylor series with $a = 0$: $\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n.$
Interval of convergence	The set of x -values for which the series converges; end-points must be tested separately.

11 Practice Problems

Practice Problems

1. Determine the intervals of concavity and all inflection points of

$$f(x) = x^4 - 4x^3.$$

2. Find the intervals on which

$$f(x) = \ln(x^2 + 1)$$

is concave up or concave down.

3. Show that $x = 0$ is not an inflection point of $f(x) = x^4$, although $f''(0) = 0$.
4. Find the Taylor polynomial of degree 4 for $f(x) = e^x$ about $x = 1$.
5. Find the Taylor series of $\sin x$ about $x = \pi$.
6. Find the Taylor series of $\frac{1}{x}$ about $x = 2$.
7. Use a known Maclaurin series to find a series for xe^{x^2} .
8. Use the Maclaurin polynomial of degree 4 for e^x to approximate $e^{0.2}$.

9. Find the Taylor series for the following functions:

(a) $\sin x$, at $x_0 = \frac{\pi}{2}$.

(b) $\ln x$, at $x_0 = 2$.

10. Expand $y = \ln x$ in powers of $x - 2$ and $y = e^{ax}$ in powers of $x - 1$.

11. Find the Maclaurin series for the functions e^{ax} and $\cos x$.

12. Find the Maclaurin polynomials p_1 , p_2 , p_3 , and p_4 for $e^x \cos x$.

13. Expand $y = \ln(x + 1)$ and $y = \frac{\sin x}{\cos x}$ in powers of x .